MATH 152 **EXAM II** NAME Solutions

Mrs. Bonny Tighe 50 points

7.7-11.7 TA_____ Wed 10/31/2018

PART II

1. (10 points) Complete the following tests

GEOMETRIC SERIES $\sum_{n=1}^{\infty} a_n (r)^{n-1}$ is convergent if $|r| < 1$

THE COMPARISON TEST Suppose $\sum_{n=1}^{\infty} a_n$ and $\sum_{n=1}^{\infty} b_n$ are series with positive terms.

i) If $\sum_{n=1}^{\infty} b_n$ is divergent and $a_n \geq b_n$ for all n, then $\sum_{n=1}^{\infty} a_n$ is also divergent

ii) If $\sum_{n=1}^{\infty} b_n$ is convergent and $a_n \leq b_n$ for all n, then $\sum_{n=1}^{\infty} a_n$ is also convergent

THE LIMIT COMPARISON TEST

Suppose that $\sum_{n=1}^{\infty} a_n$ and $\sum_{n=1}^{\infty} b_n$ are series with positive terms. If $\lim_{n\to\infty} \frac{a_n}{b_n} = c$ where c is >0 and finite, then either both series converge or both diverge.

Name the test: Test for Divergence **If** $\lim_{n\to\infty} a_n \neq 0$, then $\sum_{n=1}^{\infty} a_n$ is divergent

THE INTEGRAL TEST Suppose f is a continuous, positive, decreasing function on $[1, \infty)$ and let $a_n = f(n)$ then the series $\sum_{n=1}^{\infty} a_n$ is convergent if and only if $\int_a^{\infty} f(x)\,dx$ is convergent.

P-SERIES $\sum_{n=1}^{\infty} \frac{1}{n^p}$ is convergent if $p \geq 1$

RATIO TEST

If $\lim_{n\to\infty} \left| \frac{a_{n+1}}{a_n} \right| = L$

$L <$ 1 then $\sum a_n$ is absolutely convergent

$L >$ 1 then $\sum a_n$ is divergent

$L =$ 1 then the test is inconclusive

ROOT TEST

If $\lim_{n\to\infty} \sqrt[n]{|a_n|} = L$ and

$L <$ 1 then $\sum a_n$ is absolutely convergent

$L >$ 1 then $\sum a_n$ is divergent

$L =$ 1 then the root test is inconclusive

THE ALTERNATING SERIES TEST

If i) $b_{n+1} \leq b_n$ for all n and ii) $\lim_{n\to\infty} b_n = 0$ Then $\sum_{n=1}^{\infty} (-1)^{n-1} b_n$ is convergent

REMAINDER ESTIMATE FOR THE INTEGRAL TEST

Suppose $a_k = f(k)$ where f is a continuous, positive, decreasing function for $x \geq n$, and $\sum a_n$ is convergent. If $R_n = S - S_n$, then $\int_{n+1}^{\infty} f(k)\,dk \leq R_n \leq \int_n^{\infty} f(x)\,dk$

2. (10 points). Use the **Integral test** to determine if the series is divergent or convergent.

$$\sum_{n=1}^{\infty} \frac{1}{\sqrt{n}\, 4^{\sqrt{n}}}$$

Let $f(n) = \frac{4^{-\sqrt{n}}}{\sqrt{n}}$ which is continuous, positive and decreasing for $n \geq 1$

$$\lim_{t\to\infty} \int_1^t 4^{-\sqrt{n}}\left(\frac{1}{\sqrt{n}}\right)dn = \qquad u = -\sqrt{n}, \quad du = -\frac{1}{2\sqrt{n}}\,dn, \quad -2\sqrt{n}\,du = dn$$

$$\lim_{t\to\infty} -2\int_1^t (4^u)\,du = \lim_{t\to\infty} -2\,\frac{4^{-\sqrt{n}}}{\ln 4}\Big]_1^t$$

$$\lim_{t\to\infty} \frac{-2}{4^{\sqrt{n}}\ln 4}\Big]_1^t = \lim_{t\to\infty}\left[\frac{-2}{4^{\sqrt{t}}\ln 4} - \left(\frac{-2}{4\ln 4}\right)\right] = \frac{2}{4\ln 4}$$

which is finite so converges

So $\therefore$ Thus $\sum_{n=1}^{\infty} \frac{1}{\sqrt{n}\,4^{\sqrt{n}}}$ is also convergent

3. (10 points) Determine whether the series is convergent or divergent and if it is convergent **find the sum**.

a) $\sum_{k=1}^{\infty} \frac{1}{k^2+4k+3}$

$$\frac{A}{k+1} + \frac{B}{k+3} = \frac{1}{(k+3)(k+1)}$$

$A(k+3) + B(k+1) = 1$

$k=-1 \quad A = 2$

$k=-3 \quad B = -2$

$$\sum_{k=1}^{\infty}\left(\frac{2}{k+1} - \frac{2}{k+3}\right)$$

$$\frac{2}{2} + \frac{2}{3} + \cancel{\frac{2}{4}} + \cancel{\frac{2}{5}} + \cancel{\frac{2}{6}} \ldots$$

$$-\cancel{\frac{2}{4}} \quad -\cancel{\frac{2}{5}} \quad -\cancel{\frac{2}{6}}$$

Sum $= 1 + 2/3 = 5/3$ I found the sum so $\sum_{k=1}^{\infty} \frac{1}{k^2+4k+3}$ is convergent

b) $\sum_{k=1}^{\infty} \frac{3^{k+1}}{5^{k+2}}$ $\quad \sum_{k=1}^{\infty} \frac{3}{5^2}\left(\frac{3}{5}\right)^k$

$|r| = 3/5 < 1$ so convergent by the geometric series test. (Because it is convergent I can find the sum) $\sum_{k=1}^{\infty} ar^{k-1} = \frac{a}{1-r}$

$$\sum_{k=1}^{\infty} \frac{9}{125}\left(\frac{3}{5}\right)^{k-1} = \frac{9/125}{1-3/5}$$

$$\frac{9}{125-75} = \frac{9}{50}$$

Could use Direct Comparison Test as well

4. (10 points) Test the following series for convergence or divergence. State the test you are using and show all your work.

abs + cond

a) $\sum_{k=1}^{\infty} \frac{k-1}{\sqrt{k^5+3k^2+1}}$ $\quad \frac{k}{k^{5/2}}$

Compare to $\sum_{k=1}^{\infty} \frac{1}{k^{3/2}}$ a convergent p-series, $p = 3/2 > 1$

Limit Comparison Test

$$\lim_{k\to\infty}\left(\frac{k-1}{\sqrt{k^5+3k^2+1}} \cdot \frac{k^{3/2}}{1}\right) =$$

$$\lim_{k\to\infty}\left(\frac{k^{5/2}-k^{3/2}}{\sqrt{k^5+3k^2+1}}\right) = 1 > 0$$ and finite

So $\sum_{k=1}^{\infty} \frac{k-1}{\sqrt{k^5+3k^2+1}}$ is also Absolutely Convergent

b) $\sum_{n=1}^{\infty} \frac{(-1)^{n-1}}{\sqrt{n+2}}$

Compare to $\sum_{n=1}^{\infty} \frac{1}{\sqrt{n}}$ a divergent p-series

$b_n = \frac{1}{\sqrt{n+2}}$

$p = 1/2 \le 1$

Limit Comparison Test

$$\lim_{n\to\infty}\left(\frac{1}{\sqrt{n+2}} \cdot \frac{\sqrt{n}}{1}\right) = 1 > 0$$ + finite

So Both are Divergent

Now use Alternating Series Test

$b_n = \frac{1}{\sqrt{n+2}}$ $\quad$ i) $b_{n+1} \le b_n$

$\frac{1}{\sqrt{n+3}} \le \frac{1}{\sqrt{n+2}}$ ✓

ii) $\lim_{n\to\infty} b_n = 0$ $\quad \lim_{n\to\infty} \frac{1}{\sqrt{n+2}} = 0$

So $\sum_{n=1}^{\infty} \frac{(-1)^{n-1}}{\sqrt{n+2}}$ is Conditionally Convergent

c) $\sum_{n=1}^{\infty} \frac{(3\cdot 5\cdot 7\cdot 9\cdots(2n+1))}{e^n\, n!}$

Ratio Test

$$\lim_{n\to\infty}\left|\frac{3\cdot5\cdot7\cdot9\cdots(2n+1)(2n+3)}{e\cdot e^n\cdot(n+1)n!} \cdot \frac{e^n\, n!}{(3\cdot5\cdot7\cdots(2n+1))}\right|$$

$$\lim_{n\to\infty}\left|\frac{2n+3}{e(n+1)}\right| = \frac{2}{e} < 1$$ So Absolutely Convergent

5. (10 points) Multiple Choice: Circle the correct letter

a) Find a sequence for $1, \frac{1\cdot 5}{1\cdot 2}, \frac{1\cdot 5\cdot 9}{1\cdot 2\cdot 3}, \frac{1\cdot 5\cdot 9\cdot 13}{1\cdot 2\cdot 3\cdot 4}, \ldots$ is

$n=4$

a) $a_n = \frac{4n-3}{n!}$

b) $a_n = \frac{(1\cdot 5\cdot 9\cdots(4n-3))}{(n+1)!}$

c) $a_n = \frac{(1\cdot 5\cdot 9\cdots(4n-3))}{n!}$

d) $a_n = \frac{4n-1}{(n+1)!}$

e) None of the above

b) $\sum_{n=1}^{\infty} \frac{(-5)^{n+1}}{3^{2n}}$ equals

$$\sum_{n=1}^{\infty} -5\left(\frac{-5}{9}\right)^n = \sum_{n=1}^{\infty} \frac{25}{9}\left(\frac{-5}{9}\right)^{n-1}$$

$$\frac{25/9}{1-(-5/9)} = \frac{25}{9+5}$$

a) $\frac{-45}{14}$

b) $\frac{-45}{4}$

c) $\frac{25}{14}$

d) $\frac{-25}{4}$

e) None of the above

c) Which of the following sequences is not convergent?

a) $a_n = \frac{4n-1}{2n+1}$

b) $a_n = \frac{5n+2}{3n}$

c) $a_n = \frac{3}{n!}$

d) $a_n = \frac{\arctan n}{\sqrt{n+2}}$

e) They are all convergent

d) Which test would you use to determine the convergence of the series $\sum_{n=1}^{\infty} \frac{(-1)^{n+1}}{n^3+3n+1}$

a) Alternating Series Test
b) Integral Test
c) Limit Comparison Test
d) Ratio Test
e) None of the above tests

e) Which test would you use to determine the convergence of the series $\sum_{n=0}^{\infty} \frac{(-1)^n}{n!}$

a) Limit Comparison Test
b) Root Test
c) Integral Test
d) Ratio Test
e) None of the above
f) 6. (5 points) True False – Write the word

a) The first test you try when determining the convergence of a series is the ratio test. False

b) Given $a_n = n^2 + 2n + 1$, then $a_{n+1} = n^2 + 4n + 4$. True

c) To determine the convergence of $\sum_{n=0}^{\infty} \frac{n-1}{n^2+n^3}$ I could use the Integral Test.

False $n=0$ is negative

d) If $\lim_{n\to\infty} a_n = 0$, then the series $\sum_{n=0}^{\infty} a_n$ is convergent. False

e) If $\lim_{n\to\infty} a_n = 0$, then the sequence $\{a_n\}$ is convergent. True

MATH 152
Mrs. Bonny Tighe

EXAM II

50 points
7.7-11.7
PART I

NAME Solutions

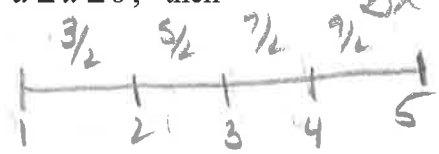
TA_____ Wed 10/31/2018

7

1. (10 points) Use (a) the Midpoint Rule (b) the Trapezoidal Rule and (c) Simpson's Rule to approximate the given integral with the specified value of n. Find the error bounds for the Midpoint rule. Suppose $|f''(x)| \le K$ *for* $a \le x \le b$, then

$$|E_M| \le \frac{K(b-a)^3}{24n^2} \qquad \int_1^5 \frac{1}{x^4}\,dx, \quad n = 4$$

$\Delta x = 1$



a) $M_4 = \frac{1}{2}\left[\frac{1}{(3/2)^4} + \frac{1}{(5/2)^4} \quad \frac{1}{(7/2)^4} \quad \frac{1}{(9/2)^4}\right]$

b) $T_4 = \frac{1}{4}\left[\frac{1}{1^4} + 2\left(\frac{1}{2^4}\right) + 2\left(\frac{1}{3^4}\right) + 2\left(\frac{1}{4^4}\right) + \frac{1}{5^4}\right]$

c) $S_4 = \frac{1}{6}\left[\frac{1}{1} + 4\left(\frac{1}{2^4}\right) + 2\left(\frac{1}{3^4}\right) + 4\left(\frac{1}{4^4}\right) + \frac{1}{5^4}\right]$

$f(x) = x^{-4}$
$f'(x) = -4x^{-5}$
$f''(x) = 20x^{-6}$
$\left|\frac{20}{x^6}\right|$

$|E_M| \le \frac{20(5-1)^3}{24(4)^2} = \frac{20(4)^3}{24(4)^2} = \frac{20(4)}{24} \le 10/3$ greatest value at $x = 1$

$K = 20$

3

How large should we take n in order to guarantee that the Midpoint approximation for $\int_1^5 \frac{1}{x^4}dx$ is accurate to within 0.001?

$K = 20$

$$\frac{20(4)^3}{24n^2} \le \frac{1}{10^3}$$

$$10^3(20)4^3 \le 24n^2$$

$$\sqrt{\frac{10^3(20)4^3}{24}} \le n$$ the next integer

2. (10 points) Find the length of the arc of the parabola $y = \frac{x^2}{4} - \frac{1}{2}\ln(x)$, $1 \le x \le 2$

dx

$L = \int_a^b \sqrt{1 + (dy/dx)^2}\, dx$

$dy/dx = \frac{1}{4}(2x) - \frac{1}{2x} = \frac{1}{2}x - \frac{1}{2x}$

$L = \int_1^2 \sqrt{1 + \frac{1}{4}x^2 - \frac{1}{2} + \frac{1}{4x^2}}\, dx$

$(dy/dx)^2 = \frac{1}{4}x^2 - 2(\frac{1}{2}x)(\frac{1}{2x}) + \frac{1}{4x^2}$

$= \frac{1}{4}x^2 - \frac{1}{2} + \frac{1}{4x^2}$

"one of those"

$= \int_1^2 \sqrt{\frac{1}{4}x^2 + \frac{1}{2} + \frac{1}{4x^2}}\, dx = \int_1^2 \sqrt{(\frac{1}{2}x + \frac{1}{2x})^2}\, dx$

$= \int_1^2 (\frac{1}{2}x + \frac{1}{2}(\frac{1}{x}))\, dx = \frac{1}{4}x^2 + \frac{1}{2}\ln|x| \Big]_1^2 =$

$(\frac{1}{4}(2)^2 + \frac{1}{2}\ln(2)) - (\frac{1}{4} + \frac{1}{2}\ln|1|) = 1 + \frac{1}{2}\ln|2| - \frac{1}{4} = 3/4 + \frac{1}{2}\ln|2|$

3. (10 points) Determine if the following improper integrals are convergent or divergent. If convergent, evaluate.

a) $\int_1^2 \frac{3x}{\sqrt{x^2-1}}\, dx$

$u = x^2 - 1$
$du = 2x\, dx$

$\lim_{t \to 1^+} \int_t^2 (x^2-1)^{-1/2}(3x)\, dx$

$\lim_{t \to 1^+} \frac{3}{2} \int u^{-1/2}\, du =$

$\lim_{t \to 1^+} \frac{3}{2} \cdot \frac{1}{1/2} u^{1/2} \Big]$

$\lim_{t \to 1^+} 3\sqrt{x^2-1} \Big]_t^2$

$\lim_{t \to 1^+} [3\sqrt{3} - 3\sqrt{t^2-1}]$

$3\sqrt{3}$

So convergent

b) $\int_0^\infty x e^{-x^2}\, dx$

$\lim_{t \to \infty} \int_0^t e^{-x^2} x\, dx$

$\lim_{t \to \infty} -\frac{1}{2} e^{-x^2} \Big]_0^t$

$\lim_{t \to \infty} [-\frac{1}{2e^{t^2}} - (-\frac{1}{2e^0})] =$

$\frac{1}{2}$ so

convergent

4. (5 points) True False – Write the word

a) The formula for Simpson's rule for approximation of a definite integral with four subdivisions is $\Delta x[f(x_0)+4f(x_1)+2f(x_2)+4f(x_3)+f(x_4)]$. False

b) This $\int_1^3 \frac{x}{\sqrt{x^3+2x-3}}dx$ is an improper integral. True

c) The arc length of the function $y=\sqrt{x}$ on the interval $2 \le y \le 3$ is $\int_2^3 \sqrt{1+\left(\frac{1}{4x}\right)}dx$.

dy

False $dy/dx = \frac{1}{2\sqrt{x}}$ $\frac{1}{4x}$

d) The approximation for Trapezoid rule for a definite integral with four subdivisions is $\Delta x/2[f(x_1)+2f(x_2)+2f(x_3)+f(x_4)]$. False

e) The arc length for the function $y=e^{3x}$ on the interval $1 \le x \le 2$ is

dx

$\int_1^2 \sqrt{1+\left(9e^{9x^2}\right)}dx$ False.

$dy/dx = 3e^{3x}$

$(dy/dx)^2 = 9e^{6x}$

5. (10 points) Multiple Choice – (2 points each) Circle the correct letter

a) When using the comparison test for the improper integral $\int_2^\infty \frac{x}{\sqrt{x^3-2x+3}}dx$ the inequality will be

$\frac{x}{x^{3/2}}$ $\frac{1}{x^{1/2}}$

a) $\frac{1}{\sqrt{x^3}} \le \frac{x}{\sqrt{x^3-2x+3}}$

b) $\frac{1}{\sqrt{x^3}} \ge \frac{x}{\sqrt{x^3-2x+3}}$

c) $\frac{1}{\sqrt{x}} \le \frac{x}{\sqrt{x^3-2x+3}}$

d) $\frac{1}{\sqrt{x}} \ge \frac{x}{\sqrt{x^3-2x+3}}$

e) None of the above

$r = x$

b) The surface area for the function $y = \ln x$ when rotated about the y-axis on the interval $2 \le y \le 3$ is:

$x = e^y$

dy

a) $2\pi \int_2^3 y\sqrt{1+e^{2y}}\,dy$

b) $2\pi \int_2^3 e^y \sqrt{1+e^{2y}}\,dy$

c) $2\pi \int_2^3 x\sqrt{1+\left(\frac{1}{x^2}\right)}\,dx$

d) $2\pi \int_2^3 \ln x\sqrt{1+\left(\frac{1}{x^2}\right)}\,dx$

e) None of the above

c) The K for the error bound for the midpoint formula for approximating $\int_1^4 \sqrt{x}\,dx$ with 4 subdivisions is:

$f'(x) = \frac{1}{2}x^{-1/2}$

$f''(x) = -\frac{1}{4}x^{-3/2} = \left(-\frac{1}{4x^{3/2}}\right)$

a) $K = -\frac{1}{4}$

b) $K = \frac{1}{2}$

c) $K = \frac{1}{4}$

d) $K = \frac{1}{32}$

e) None of the above

d) Which of the following is not an improper integral?

a) $\int_{-2}^{5} \frac{1}{\sqrt{x+2}}\,dx$

b) $\int_1^3 \frac{x}{x^2+3x-4}\,dx$

c) $\int_1^3 \frac{x}{x^2-3x+4}\,dx$

d) $\int_0^1 \frac{1}{\ln x}\,dx$

e) They are all improper integrals

e) Represent $0.4\overline{7}$ as a series

a) $\frac{4}{10}+\sum_{n=0}^{\infty}\left(\frac{7}{10}\right)^n$

b) $\frac{4}{10}+\sum_{n=0}^{\infty}\frac{7}{10}\left(\frac{1}{10}\right)^n$

c) $\frac{4}{10}+\sum_{n=0}^{\infty}\frac{7}{10}\left(\frac{1}{100}\right)^n$

d) $\frac{4}{10}+\sum_{n=0}^{\infty}7\left(\frac{1}{10}\right)^n$

e) None of the above

$$\frac{4}{10}+\sum_{n=0}^{\infty}\frac{7}{100}\left(\frac{1}{10}\right)^n$$

6. (10 points) The given curve is rotated about the y-axis. Find the area of the resulting surface. $S=2\pi\int radius\ (arclength)$ $\quad x=\sqrt{1+e^y},\ \ 0\le y\le 3$

$r=x$

dy

$$S=2\pi\int_0^3\sqrt{1+e^y}\sqrt{1+\frac{e^{2y}}{4(1+e^y)}}\,dy$$

$$dx/dy=\frac{1}{2}(1+e^y)^{-1/2}e^y=\frac{e^y}{2\sqrt{1+e^y}}$$

$$=2\pi\int_0^3\sqrt{1+e^y}\sqrt{\frac{4+4e^y+e^{2y}}{4(1+e^y)}}\,dy$$

$$2\pi\int_0^3\cancel{\sqrt{1+e^y}}\,\frac{\sqrt{(e^y+2)^2}}{2\cancel{\sqrt{1+e^y}}}\,dy$$

$$\pi\int_0^3(e^y+2)\,dy=\pi\left[e^y+2y\right]_0^3=$$

$$\pi\left[(e^3+6)-(e^0+0)\right]=\pi\left[e^3+5\right]$$

MATH 152 **EXAM IIA** NAME Solutions

Mrs. Bonny Tighe 50 points

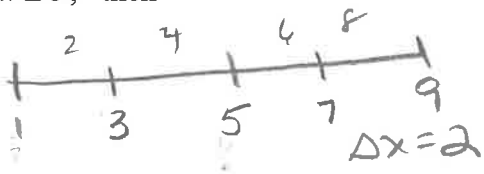
7.7-11.7 TA_____ Wed 10/31/2018

PART I

1. (10 points) Use (a) the Midpoint Rule (b) the Trapezoidal Rule and (c) Simpson's Rule to approximate the given integral with the specified value of n. Find the error bounds for the Midpoint rule. Suppose $|f''(x)| \le K$ *for* $a \le x \le b$, then

7

$$|E_M| \le \frac{K(b-a)^3}{24n^2} \qquad \int_1^9 \frac{1}{x^2}\,dx, \quad n=4$$



a) $M_4 = 2\left[\frac{1}{2^2}+\frac{1}{4^2}+\frac{1}{6^2}+\frac{1}{8^2}\right]$

b) $T_4 = 1\left[\frac{1}{1^2}+2\left(\frac{1}{3^2}\right)+2\left(\frac{1}{5^2}\right)+2\left(\frac{1}{7^2}\right)+\frac{1}{9^2}\right]$

c) $S_4 = \frac{2}{3}\left[\frac{1}{1^2}+4\left(\frac{1}{3^2}\right)+2\left(\frac{1}{5^2}\right)+4\left(\frac{1}{7^2}\right)+\frac{1}{9^2}\right]$

$f(x) = x^{-2}$

$f'(x) = -2x^{-3}$

$f''(x) = 6x^{-4}$

$\left|\frac{6}{x^4}\right|$

largest when $x=1$

$$|E_M| \le \frac{6(9-1)^3}{24(4)^2} \le \frac{\cancel{6}(8)^3}{\cancel{24}_{4}(16)} \le \frac{8\cdot8\cdot8}{8\cdot8} \le 8$$

3

How large should we take n in order to guarantee that the Midpoint approximation for $\int_1^9 \frac{1}{x^2}dx$ is accurate to within 0.001?

$$\frac{\cancel{6}(8)^3}{\cancel{24}_{4}\,n^2} \le \frac{1}{10^3} \qquad \frac{8^3}{4n^2} \le \frac{1}{10^3}$$

$$\frac{8^3(10^3)}{4} \le n^2$$

$$\sqrt{\frac{8^3\,10^3}{4}} \le n$$ the next integer

$r = y$

2. (10 points) The given curve is rotated about the x-axis. Find the area of the resulting surface. $S = 2\pi \int_a^b radius\, arclength$ $\quad y = \sqrt{6+x},\ 3 \le x \le 4$

dx

$$dy/dx = \frac{1}{2}(6+x)^{-1/2}(+1) = \frac{+1}{2\sqrt{6+x}}$$

$$S = 2\pi \int_3^4 \sqrt{6+x}\sqrt{1+\frac{1}{4(6+x)}}\,dx$$

$$S = 2\pi \int_3^4 \sqrt{6+x}\sqrt{\frac{4(6+x)+1}{4(6+x)}}\,dx =$$

$$S = 2\pi \int_3^4 \cancel{\sqrt{6+x}}\,\frac{\sqrt{25+4x}}{2\cancel{\sqrt{6+x}}}\,dx = \pi \int_4^3 \sqrt{25+4x}\,dx$$

$$\pi \frac{1}{4}\left(\frac{1}{3/2}(25+4x)^{3/2}\right)\Big]_3^4 = \frac{\pi}{4}\cdot\frac{2}{3}(25+4x)^{3/2}\Big]_3^4$$

$$\frac{\pi}{6}\left[(25+16)^{3/2} - (25+12)^{3/2}\right] \text{ or } \frac{\pi}{6}\left[41^{3/2} - 37^{3/2}\right]$$

3. (10 points) Determine if the following improper integrals are convergent or divergent. If convergent, evaluate.

a) $\int_1^\infty \cancel{\frac{1}{(3x+2)^2}}\,dx$ $\quad \int_1^\infty xe^{-x}\,dx$

$$\lim_{t\to\infty} \int_1^t x e^{-x}\,dx$$

$u = x \quad dv = e^{-x}dx$
$du = dx \quad v = -e^{-x}$

$$\lim_{t\to\infty}\left[-xe^{-x} - \int_1^t -e^{-x}\,dx\right] =$$

$$\lim_{t\to\infty}\left[-\frac{x}{e^x} - \frac{1}{e^x}\right]_1^t$$

$$\lim_{t\to\infty}\left[-\frac{t}{e^t} - \frac{1}{e^t} - \left(-\frac{1}{e} - \frac{1}{e}\right)\right]$$

$$0 \quad 0 \quad + \frac{2}{e} = \frac{2}{e}$$

So convergent

b) $\int_1^3 \frac{x}{\sqrt{x^2-1}}\,dx$

$$\lim_{t\to 1^+} \int_1^3 x(x^2-1)^{-1/2}\,dx$$

$$\lim_{t\to 1^+} \frac{1}{2}\cdot\frac{1}{1/2}(x^2-1)^{1/2}\Big]_1^3$$

$$\lim_{t\to 1^+} \sqrt{x^2-1}\Big]_1^3 =$$

$$\lim_{t\to 1^+}\left(\sqrt{9-1} - \sqrt{t^2-1}\right)$$

$$\sqrt{8} - 0 = \sqrt{8}$$

So convergent

4. (10 points) Find the length of the arc of the parabola

$$x = \frac{y^2}{2} - \frac{1}{4}\ln y \quad on \quad 1 \le y \le 2 \quad dy$$

$$x = \frac{1}{2}y^2 - \frac{1}{4}\ln y$$

$$dx/dy = y - \frac{1}{4}\cdot\frac{1}{y}$$

$$(dx/dy)^2 = y^2 - 2(y)\left(\frac{1}{4y}\right) + \frac{1}{16y^2}$$

$$y^2 - \frac{1}{2} + \frac{1}{16y^2}$$

one of those

$$L = \int_a^b \sqrt{1 + (dx/dy)^2}\, dy$$

$$L = \int_1^2 \sqrt{1 + y^2 - \frac{1}{2} + \frac{1}{16y^2}}\, dy$$

$$= \int_1^2 \sqrt{y^2 + \frac{1}{2} + \frac{1}{16y^2}}\, dy$$

$$L = \int_1^2 \sqrt{\left(y + \frac{1}{4y}\right)^2}\, dy = \int_1^2 \left(y + \frac{1}{4}\cdot\frac{1}{y}\right) dy$$

$$\frac{1}{2}y^2 + \frac{1}{4}\ln|y| \Big]_1^2 =$$

$$\left(\frac{1}{2}(4) + \frac{1}{2}\ln|2|\right) - \left(\frac{1}{2} + \frac{1}{4}\ln|1|\right)$$

$$2 + \frac{1}{2}\ln|2| - \frac{1}{2} = 3/2 + \frac{1}{2}\ln|2|$$

5. (10 points) Multiple Choice – (2 points each) Circle the correct lettera) When using the comparison test for the improper integral $\int_2^\infty \frac{x}{\sqrt{x^3 - 2x + 3}}dx$ the inequality will be

a) $\frac{1}{\sqrt{x^3}} \le \frac{x}{\sqrt{x^3 - 2x + 3}}$

b) $\frac{1}{\sqrt{x^3}} \ge \frac{x}{\sqrt{x^3 - 2x + 3}}$

c) $\frac{1}{\sqrt{x}} \le \frac{x}{\sqrt{x^3 - 2x + 3}}$

d) $\frac{1}{\sqrt{x}} \ge \frac{x}{\sqrt{x^3 - 2x + 3}}$

e) None of the above

$$\frac{x}{x^{3/2}} \qquad \frac{1}{x^{1/2}} \text{ Diverges}$$

$r = x$

b) The surface area for the function $y = \ln x$ when rotated about the y-axis on the interval $2 \le y \le 3$ is: dy $\quad x = e^y$

a) $2\pi \int_2^3 y\sqrt{1+e^{2y}}\,dy$

b) $2\pi \int_2^3 e^y \sqrt{1+e^{2y}}\,dy$

c) $2\pi \int_2^3 x\sqrt{1+\left(\frac{1}{x^2}\right)}\,dx$

d) $2\pi \int_2^3 \ln x\sqrt{1+\left(\frac{1}{x^2}\right)}\,dx$

e) None of the above

c) The K for the error bound for the midpoint formula for approximating $\int_1^4 \sqrt{x}\,dx$ with 4 subdivisions is:

a) $K = -\frac{1}{4}$

b) $K = \frac{1}{2}$

c) $K = \frac{1}{4}$

d) $K = \frac{1}{32}$

e) None of the above

d) Which of the following is not an improper integral?

a) $\int_{-2}^{5} \frac{1}{\sqrt{x+2}}\,dx$

b) $\int_1^3 \frac{x}{x^2+3x-4}\,dx$

c) $\int_1^3 \frac{x}{x^2-3x+4}\,dx$

d) $\int_0^1 \frac{1}{\ln x}\,dx$

e) They are all improper integrals

e) Represent $0.4\overline{7}$ as a series

a) $\frac{4}{10}+\sum_{n=0}^{\infty}\left(\frac{7}{10}\right)^n$

b) $\frac{4}{10}+\sum_{n=0}^{\infty}\frac{7}{10}\left(\frac{1}{10}\right)^n$

c) $\frac{4}{10}+\sum_{n=0}^{\infty}\frac{7}{10}\left(\frac{1}{100}\right)^n$

d) $\frac{4}{10}+\sum_{n=0}^{\infty}7\left(\frac{1}{10}\right)^n$

e) None of the above

6. (5 points) True False – Write the word

a) The formula for Simpson's rule for approximation of a definite integral with four subdivisions is $\Delta x[f(x_0)+4f(x_1)+2f(x_2)+4f(x_3)+f(x_4)]$. False

b) This $\int_1^3 \frac{x}{\sqrt{x^3+2x-3}}dx$ is an improper integral. True

c) The arc length of the function $y=\sqrt{x}$ on the interval $2\le y\le 3$ is $\int_2^3 \sqrt{1+\left(\frac{1}{4x}\right)}dx$.

dy

False

d) The approximation for Trapezoid rule for a definite integral with four subdivisions is $\Delta x/2\,[f(x_1)+2f(x_2)+2f(x_3)+f(x_4)]$. False

e) The arc length for the function $y=e^{3x}$ on the interval $1\le x\le 2$ is

$\int_1^2 \sqrt{1+\left(9e^{9x^2}\right)}dx$ False

MATH 152 **EXAM IIA** NAME Solution

Mrs. Bonny Tighe 50 points

7.7-11.7 TA_____ Wed 10/31/2018

PART II

1. (10 points) Complete the following

GEOMETRIC SERIES $\sum_{n=1}^{\infty} a_n (r)^{n-1}$ is divergent if $|r| \geq 1$

THE COMPARISON TEST Suppose $\sum_{n=1}^{\infty} a_n$ and $\sum_{n=1}^{\infty} b_n$ are series with positive terms.

i) If $\sum_{n=1}^{\infty} b_n$ is convergent and $a_n \leq b_n$ for all n, then $\sum_{n=1}^{\infty} a_n$ is also convergent

ii) If $\sum_{n=1}^{\infty} b_n$ is divergent and $a_n \geq b_n$ for all n, then $\sum_{n=1}^{\infty} a_n$ is also divergent

THE LIMIT COMPARISON TEST

Suppose that $\sum_{n=1}^{\infty} a_n$ and $\sum_{n=1}^{\infty} b_n$ are series with positive terms . If $\lim_{n\to\infty} \frac{a_n}{b_n} = c$ where c is > 0 and finite, then either both series converge or both diverge.

TEST FOR DIVERGENCE **If** $\lim_{n\to\infty} a_n \neq 0$, then $\sum_{n=1}^{\infty} a_n$ is divergent

THE INTEGRAL TEST Suppose f is a continuous, positive, decreasing function on $[1, \infty)$ and let $a_n = f(n)$ then the series $\sum_{n=1}^{\infty} a_n$ is convergent if and only if $\int_1^{\infty} f(n)\,dn$ is convergent.

P-SERIES $\sum_{n=1}^{\infty} \frac{1}{n^p}$ is convergent if $p > 1$

RATIO TEST

If $\lim_{n\to\infty} \left| \frac{a_{n+1}}{a_n} \right| = L$ $\begin{cases} L < 1 \text{ then } \sum a_n \text{ is absolutely convergent} \\ L > 1 \text{ then } \sum a_n \text{ is divergent} \\ L = 1 \text{ then the test is inconclusive} \end{cases}$

ROOT TEST

If $\lim_{n\to\infty} \sqrt[n]{|a_n|} = L$ and $\begin{cases} L < 1 \text{ then } \sum a_n \text{ is absolutely convergent} \\ L > 1 \text{ then } \sum a_n \text{ is divergent} \\ L = 1 \text{ then the root test is inconclusive} \end{cases}$

THE ALTERNATING SERIES TEST

If i) is $b_{n+1} \leq b_n$ for all n and ii) $\lim_{n\to\infty} b_n = 0$ Then $\sum_{n=1}^{\infty} (-1)^{n-1} b_n$ is convergent

REMAINDER ESTIMATE FOR THE INTEGRAL TEST

Suppose $a_k = f(k)$ where f is a continuous, positive, decreasing function for $x \geq n$, and $\sum a_n$ is convergent . If $R_n = S - S_n$, then $\int_{n+1}^{\infty} f(k)\,dk \leq R_n \leq \int_n^{\infty} f(k)\,dk$

2. (10 points). Use the **Integral test** to determine if the series is divergent or convergent.

$$\sum_{n=0}^{\infty} \frac{e^n}{e^{2n}+1}$$

$f(n) = \frac{e^n}{e^{2n}+1}$ is continuous, decreasing and positive for $n \geq 0$

$$\lim_{t\to\infty} \int_0^t \frac{e^n}{e^{2n}+1}\,dn = \lim_{t\to\infty} \tan^{-1}(e^n)\Big]_0^t =$$

$$\lim_{t\to\infty} \left[\tan^{-1}(e^t) - \tan^{-1}(e^0)\right] = \pi/2 - \pi/4 = \pi/4$$

So convergent

So $\sum_{n=0}^{\infty} \frac{e^n}{e^{2n}+1}$ is also Converges

$$\cancel{\int_{11}^{\infty} \frac{e^n}{e^{2n}+1} \leq R_{10} \leq \int_{10}^{\infty} \frac{e^n}{e^{2n}+1}\,dn \quad \tan^{-1}}$$

$$\cancel{\tan^{-1}(e^n)\Big]_{11}^{\infty} \leq R_{10} \leq \tan^{-1}(e^n)\Big]_{10}^{\infty}}$$

$$\cancel{\pi/2 - \tan^{-1}(e^{11}) \leq R_{10} \leq \pi/2 - \tan^{-1}(e^{10})}$$

3. (10 points) Multiple Choice: Circle the correct letter

a) Find a sequence for $1, \frac{1\cdot 5}{1\cdot 2}, \frac{1\cdot 5\cdot 9}{1\cdot 2\cdot 3}, \frac{1\cdot 5\cdot 9\cdot 13}{1\cdot 2\cdot 3\cdot 4}, \ldots$ is

$n=4$

a) $a_n = \frac{4n-3}{n!}$

b) $a_n = \frac{(1\cdot 5\cdot 9\cdots(4n-3))}{(n+1)!}$

c) $a_n = \frac{(1\cdot 5\cdot 9\cdots(4n-3))}{n!}$

d) $a_n = \frac{4n-1}{(n+1)!}$

e) None of the above

b) $\sum_{n=1}^{\infty} \frac{(-5)^{n+1}}{3^{2n}}$ equals

$\sum_{n=1}^{\infty} -5\left(\frac{-5}{9}\right)^n$ $\sum_{n=1}^{\infty} \frac{25}{9}\left(\frac{-5}{9}\right)^n$

$S_\infty = \frac{25/9}{1-(-5/9)} = \frac{25}{9+5}$

a) $\frac{-45}{14}$

b) $\frac{-45}{4}$

c) $\frac{25}{14}$

d) $\frac{-25}{4}$

e) None of the above

c) Which of the following sequences is not convergent?

a) $a_n = \frac{4n-1}{2n+1}$

b) $a_n = \frac{5^x+2}{3^x}$

c) $a_n = \frac{3}{n!}$

d) $a_n = \frac{\arctan n}{\sqrt{n+2}}$

e) They are all convergent

d) Which test would you use to determine the convergence of the series $\sum_{n=1}^{\infty} \frac{(-1)^{n+1}}{n^3+3n+1}$

a) Alternating Series Test
b) Integral Test
c) Limit Comparison Test
d) Ratio Test
e) None of the above tests

e) Which test would you use to determine the convergence of the series $\sum_{n=0}^{\infty} \frac{(-1)^n}{n!}$

a) Limit Comparison Test
b) Root Test
c) Integral Test
d) Ratio Test
e) None of the above

4. (5 points) True False – Write the word

a) The first test you try when determining the convergence of a series is the ratio test. False

b) Given $a_n = n^2 + 2n + 1$, then $a_{n+1} = n^2 + 4n + 4$. True

c) To determine the convergence of $\sum_{n=0}^{\infty} \frac{n-1}{n^2+n^3}$ I could use the Integral Test.

False

d) If $\lim_{n\to\infty} a_n = 0$, then the series $\sum_{n=0}^{\infty} a_n$ is convergent. False

e) If $\lim_{n\to\infty} a_n = 0$, then the sequence $\{a_n\}$ is convergent. True

5. (10 points) Determine whether the series is convergent or divergent and if it is convergent **find the sum**.

a) $\sum_{k=1}^{\infty} \frac{1}{k^2+3k}$

$\frac{A}{k} + \frac{B}{k+3} = \frac{1}{k(k+3)}$

$A(k+3) + Bk = 1$

$k=0 \quad A = 1/3$

$k=-3 \quad B = -1/3$

$\sum_{k=1}^{\infty} \left(\frac{1/3}{k} - \frac{1/3}{k+3}\right)$

$\frac{1/3}{1} + \frac{1/3}{2} + \frac{1/3}{3} + \cancel{\frac{1/3}{4}} + \cancel{\frac{1/3}{5}} + \cancel{\frac{1/3}{6}} \cdots$

$\cancel{-\frac{1/3}{4}} \quad \cancel{-\frac{1/3}{5}} \quad \cancel{-\frac{1/3}{6}}$

Sum $= \frac{1}{3} + \frac{1}{6} + \frac{1}{9}$

I found the sum so the series $\sum_{k=1}^{\infty} \frac{1}{k^2+3k}$ is Convergent

b) $\sum_{k=0}^{\infty} \frac{3^k}{5^{k+1}}$

$\sum_{k=1}^{\infty} \frac{3 \cdot 3^{k-1}}{5^2 \cdot 5^{k-1}}$

$\sum_{k=1}^{\infty} \frac{3}{25}\left(\frac{3}{5}\right)^{k-1}$

$r = |3/5| < 1$

So convergent geometric series

So Sum $= \frac{a}{1-r}$

$\frac{\frac{3}{25}}{1-3/5} = \frac{3}{25-15} = 3/10$

6. (10 points) Test the following series for absolute convergence, conditional convergence or divergence. State the test you are using and show all your work.

a) $\sum_{k=1}^{\infty} \frac{k+1}{\sqrt{k^5+3k^2+1}}$

$\frac{k}{k^{5/2}} = \frac{1}{k^{3/2}}$

Compare to $\sum_{k=1}^{\infty} \frac{1}{k^{3/2}}$ a convergent p-series $p = 3/2 > 1$

Limit Comparison Test

$\lim_{k\to\infty}\left(\frac{k+1}{\sqrt{k^5+3k^2+1}} \cdot \frac{k^{3/2}}{1}\right) = 1 > 0$ and finite

So $\sum_{k=1}^{\infty} \frac{k+1}{\sqrt{k^5+3k^2+1}}$ is also

Absolutely Converges

b) $\sum_{n=1}^{\infty} \frac{(-1)^n}{1+2n}$ $b_n = \frac{1}{1+2n}$

Compare to $\sum_{n=1}^{\infty} \frac{1}{n}$ a divergent p-series $p = 1 \le 1$

Limit Comparison Test

$\lim_{n\to\infty}\left(\frac{1}{1+2n} \cdot \frac{n}{1}\right) = \frac{1}{2} > 0$ and finite

$\sum_{n=1}^{\infty} \frac{1}{1+2n}$ is Divergent

Now use Alternating Series test i) $\frac{1}{1+2(n+1)} \le \frac{1}{1+2n}$ ✓

ii) $\lim_{n\to\infty} \frac{1}{1+2n} = 0$

So Conditionally Converges

c) $\sum_{n=1}^{\infty} \frac{1\cdot 4\cdot 7\cdot 10\cdots(3n-2)}{5^n n!}$

Ratio Test

$\lim_{n\to\infty}\left|\frac{1\cdot4\cdot7\cdots(3n-2)(3n+1)}{5\cdot 5^n (n+1)\cancel{n!}} \cdot \frac{\cancel{5^n}\cancel{n!}}{1\cdot4\cdot7\cdots(3n-2)}\right|$

$\lim_{n\to\infty}\left|\frac{3n+1}{5(n+1)}\right| = 3/5 < 1$

So Absolutely Converges